

Comparison Report: RNN and BERT Architectures

Introduction

This project compares four deep learning models—Simple RNN, LSTM, GRU and BERT—for financial news sentiment classification. The aim was to classify financial news headlines into Bearish, Bullish and Neutral sentiments. Each model was trained using the same dataset and evaluated using Accuracy and Macro F1-score so that their performances could be compared fairly.

Dataset and Preprocessing

The dataset used in this project contains financial news headlines together with their corresponding sentiment labels. Each news headline belongs to one of three classes: Bearish, Bullish or Neutral. The dataset was first checked for missing or empty values to ensure good data quality before training. The data was then divided into training and validation sets so that the models could be evaluated fairly on unseen data. For the Simple RNN, LSTM and GRU models, a vocabulary was created from the training data and every word was converted into a numerical index. Words that were not present in the vocabulary were replaced using the **<UNK>** token. All sequences were padded using the **<PAD>** token so that every input had the same length before training. For the BERT model, the pretrained BERT tokenizer was used to generate input IDs and attention masks instead of manually creating a vocabulary. Class weights were applied during training to reduce the effect of class imbalance. Finally, all processed data was converted into tensors and loaded using PyTorch DataLoaders, allowing the models to process the dataset efficiently in batches.

Performance Comparison

Model	Accuracy	Macro F1
Simple RNN	43.93%	38.24%
LSTM	77.51%	66.43%
GRU	79.69%	70.12%
BERT	86.77%	83.03%

How the Models Worked Differently

Simple RNN: The Simple RNN reads the sentence one word after another and passes information to the next word. It is the simplest model, but it often forgets important information from the beginning of long sentences. This is why it achieved the lowest Accuracy (43.93%) and Macro F1-score (38.24%).

LSTM: LSTM improves the Simple RNN by adding a memory cell and gates. These gates help the model decide what information should be remembered and what should be forgotten. Because of this, LSTM captured sentence meaning much better and improved the Accuracy to 77.51%.

GRU: GRU also uses memory gates but has a simpler design than LSTM with fewer parameters. It trained faster while still remembering important information effectively. In this project it slightly outperformed LSTM, achieving 79.69% Accuracy and a Macro F1-score of 70.12%, making it the best RNN model.

BERT: Unlike the RNN models, BERT is a transformer model. It reads the entire sentence together instead of one word at a time and was pretrained on a very large text corpus. These advantages helped BERT achieve the best Accuracy (86.77%) and Macro F1-score (83.03%).

Why BERT Was Selected

BERT was selected as the final model because it consistently produced the highest Accuracy and Macro F1-score among all the models tested. It gave better predictions for all three sentiment classes and understood the meaning of complete financial news headlines more effectively than the RNN-based models. Since it achieved the best overall performance, it was chosen for deployment in the Streamlit application.

Streamlit Application

The Streamlit application allows users to enter a financial news headline and instantly predicts whether the sentiment is Bearish, Bullish or Neutral. It also displays the prediction probabilities using a bar chart, making the application simple, interactive and user-friendly for real-time sentiment prediction.

Conclusion

The results show a clear improvement from Simple RNN to LSTM, then to GRU, and finally to BERT. Among the RNN models, GRU performed the best, while BERT achieved the highest overall Accuracy and Macro F1-score. Therefore, BERT was selected as the final model for deployment because it delivered the most accurate and balanced predictions.